Muhammet Celal Gokalp

Systems & Validation Engineer | L4 Autonomy, Steering / Braking Features & HiL / Vehicle Test

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linkedin.com/in/celal-gökalp-4616ab122 | Full UK driving licence (commutable to Banbury) | Immediately available

PROFESSIONAL SUMMARY

Mechanical and automotive engineer (BSc Mechanical; MSc Sustainable Automotive, WMG Warwick) with 8 years bridging physical vehicle systems and autonomous driving: 2 years on-site at Jaguar Land Rover with AVL managing features, requirements, Tier 1 suppliers and dSPACE HiL / vehicle testing, followed by 5 years on an L4 automated driving stack at FiveAI / Bosch. My strengths are on the systems integration, requirements, supplier coordination, steering/braking feature validation, CAN signal analysis and L4 autonomy side of this role. I have not designed production steer-by-wire or brake-by-wire mechanical hardware in CATIA or released 2D drawings with ASME Y14.5 GD&T (my CAD work is in SolidWorks), and with 8 years of experience I am open to the level Zoox finds appropriate. Based in Coventry (commutable to Banbury) and immediately available.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – 11/2026

FiveAI / Bosch — Cambridge, United Kingdom

- **L4 steering & braking KPIs:** Turned UN R79 (automatic steering / lane change), Euro NCAP and customer requirements into KPIs and pass/fail criteria (lateral acceleration, longitudinal comfort accel/decel limits, road curvature metrics) across 1,000+ hours of test data; identified a drop in lane-change success above a curvature threshold that led to a steering controller fix.
- **Scenario validation & tooling:** Wrote 120+ parameterised OpenSCENARIO 2 scenarios generating thousands of test cases, built a GitHub Copilot helper that cut scenario creation from ~1 day to ~1 hour, and built Python tools and pipelines into dashboards.
- **Cross-functional & partner leadership:** Led 5 engineers for a year on L4 scenario validation; founding member of the Simulation EcoSystem solutions team on the Volkswagen Group ADA project, diagnosing errors across APIs, maps and simulation tools.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- **OEM & Tier 1 supplier management:** Based on-site at Jaguar Land Rover for 2 years as feature owner for Rear Cross Traffic Alert (with braking request), Blind Spot Monitor and Rear Pre-Crash; main technical link between JLR, AVL and the Tier 1 radar supplier.
- **Steering torque PID & ViL correlation (AVL DrivingCube):** Built Simulink / IPG CarMaker scenarios on circles of parameterised radius to find the steering-wheel torque needed at each radius. Fitted an external GNSS for position ground truth, captured vehicle response during airstrip steering-torque injection tests and HORIBA MIRA circuit runs, and ran parameter sweeps to calibrate the simulation steering torque PID controller.
- **Requirements & functional safety (steering & longitudinal features):** Wrote use cases, state diagrams, module input/output signal lists and high- to low-level requirements in DOORS and MKS Integrity (ASPICE V-model), including requirements for Lane Keep Assist, Lane Change Assist, Highway Assist and ACC. Wrote ISO 26262 safety requirements for a lane change feature and took part in its FMEA.
- **dSPACE HiL, vehicle testing & CAN analysis:** Executed validation across dSPACE HiL and ViL rigs, vehicles and the proving ground using Vector CANoe and CANape. When RCTA braking/alerting passed on the HiL rig but fired late on the track at a 45° approach, synchronised camera, CAN and radar logs in CANoe, traced the delay to the edge of the radar's field of view, and verified the supplier's calibration fix on site.
- **Electromechanical & vehicle fault-finding:** Inspected ~20 returned range-extender generators at the LEVC factory (multimeter phase-to-phase resistance and continuity checks, bench-powering the control unit to download supplier logs). Checked test-vehicle control-arm fasteners with a torque wrench, and diagnosed vehicle module faults via DTCs, fuse checks and a 12 V bench supply.

Mechanical Engineering Intern, TEMSA Global, 2014: electric bus powertrain power and torque calculations in MATLAB / Simulink.

EDUCATION

MSc Sustainable Automotive Engineering, WMG, University of Warwick, 2017 – 2018 (2:1 Merit). Dissertation: designed a test to find the thermal model variables of an electric motor stator (type-K thermocouple placement, enclosure, method; MotorCAD thermal network, Portunus, Simulink).

BSc Mechanical Engineering, Ege University, Izmir, 2011 – 2017. Dissertation: team project to design and build a 6-DOF Stewart motion platform (SolidWorks 3D CAD, MATLAB state-space model, Arduino); wired the 6 servo actuators, wrote Arduino code to command set angles, and plotted encoder signals.

SKILLS & HANDS-ON EXPERIENCE

Testing, CAN & rigs	dSPACE HiL and ViL rigs, AVL DrivingCube (ViL dyno test bed), proving ground & vehicle testing, Vector CANoe and CANape, CAN signal analysis, GNSS ground truth, multimeter & 12 V bench supply (all at AVL)
Systems & safety	Technical requirements authoring & decomposition, module signal lists, DOORS, MKS Integrity, RTC / RQM, ASPICE V-model, ISO 26262 & FMEA, SOTIF (trained) (all at AVL); Jira, Confluence, UN R79, Euro NCAP
Software & modelling	MATLAB / Simulink, IPG CarMaker, MotorCAD, SolidWorks (BSc), OpenSCENARIO 2, Python (tools built with AI help), Linux (5 years), Git, CI pipelines; C++ (reading). Not yet used: CATIA / 3DX, Polarion, ASME Y14.5 GD&T
Hands-on & details	Repairs own cars (multimeter fuse probing, torque wrench, calipers, micrometers), soldered motor cables to a PCB, builds PCs. British citizen, full UK driving licence, based in Coventry (commutable to Banbury), immediately available